# Chapter 2 – Bayes' Rule

DATA 335 – Statistical Modeling – Winter 2026

# Bayes' Rule

Let $A$ and $B$ be events. Then:

$$P(B \mid A) = \frac{P(A \mid B)P(B)}{P(A)}$$

# The law of total probability

Suppose that events $B_1, \ldots, B_n$ partition the sample space $\Omega$:

$$\Omega = \bigcup_{i=1}^{n} B_i, \qquad B_i \cap B_j = \varnothing \quad (i \neq j).$$

The *law of total probability*:

$$P(A) = \sum_{i=1}^{n} P(A \mid B_i) P(B_i).$$

Substituting for the denominator in Bayes' Rule gives:

$$P(B \mid A) = \frac{P(A \mid B) P(B)}{\sum_{i=1}^{n} P(A \mid B_i) P(B_i)}.$$

# Example: Fake news

Given:

- 2 news articles in 5 are fake.
- 4 in 15 fake news articles have an exclamation point in the title.
- 1 in 45 real news articles have an exclamation point in the title.

What is the probability that an article with an exclamation point in the title is fake?

Let $A$ be the event *the article contains an exclamation point in its title*.

Let $B_1$ be the event *the article is fake news*.

Let $B_2$ be the event *the article is real news*.

Since an article is either real or fake but not both,

$$B_1 \cup B_2 = \Omega \qquad \text{and} \qquad B_1 \cap B_2 = \varnothing.$$

Since 2 in 5 news articles are fake,

$$P(B_1) = \frac{2}{5} \qquad \text{and} \qquad P(B_2) = 1 - \frac{2}{5} = \frac{3}{5}$$

Since 4 in 15 (resp., 1 in 45) fake (resp., real) articles have an exclamation point in the title,

$$P(A \mid B_1) = \frac{4}{15} \qquad \text{and} \qquad P(A \mid B_2) = \frac{1}{45}.$$

Apply Bayes' Rule to compute the probability that an article with an exclamation point in its title is fake news:

$$
\begin{aligned}
P(B_1 \mid A) &= \frac{P(A \mid B_1)P(B_1)}{P(A \mid B_1)P(B_1) + P(A \mid B_2)P(B_2)} \\
&= \frac{\frac{4}{15} \times \frac{2}{5}}{\frac{4}{15} \times \frac{2}{5} + \frac{1}{45} \times \frac{3}{5}} \\
&= \frac{8}{9}.
\end{aligned}
$$

Thus, 8 in 9 articles with exclamation points in their titles are fake news.

# Exercise: Vampirism

A new test for vampirism has the following characteristics:

- A vampire tests positive for vampirism 95% of the time.
- A mortal tests positive for vampirism 1% of the time.

Given that vampires make up 0.1% of the population, what's the probability that someone who tests positive for vampirism is, in fact, a vampire?

$$
\begin{aligned}
&P(\text{vampire} \mid +) \\
&= \frac{P(+ \mid \text{vampire})P(\text{vampire})}{P(+ \mid \text{vampire})P(\text{vampire}) + P(+ \mid \text{mortal})P(\text{mortal})} \\
&= \frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.01 \times 0.999} \\
&= 0.087
\end{aligned}
$$

Thus, an individual that tests positive for vampirism has an 8.7% chance of being a vampire. Said differently, 91.3% of positive test results are *false positives*.

# Example: Pop versus Soda

- Men's heights in inches are normally distributed with mean 70 and standard deviation 3.
- Women's heights in inches are normally distributed with mean 64.5 and standard deviation 2.5.

Assuming that men and women are equally represented in the population, what's the probability that a person $> 68$ inches tall is a man?

We introduce random variables $X$ for height and $Y$ for sex, the latter coded as 0 for female and 1 for male.

Note that the events $Y = 0$ and $Y = 1$ are mutually exclusive and exhaust all possibilities.

By Bayes' Rule,

$$
\begin{aligned}
&P(Y = 1 \mid X > 68) \\
&= \frac{P(X > 68 \mid Y = 1)P(Y = 1)}{P(X > 68 \mid Y = 0)P(Y = 0) + P(X > 68 \mid Y = 1)P(Y = 1)}.
\end{aligned}
$$

Since men and women are equally represented in the population,

$$P(Y = 0) = 0.5 = P(Y = 1).$$

The conditional probabilities $P(X > x \mid Y = y)$ are values of *survival functions* of the normal distributions.

The survival function $\mathrm{sf}(x \mid \mu, \sigma)$ of the normal distribution with mean $\mu$ and standard deviation $\sigma$ is:

$$\mathrm{sf}(x \mid \mu, \sigma) = P(X > x), \qquad X \sim N(\mu, \sigma)$$

Thus,

$$P(X > 68 \mid Y = 0) = \mathrm{sf}(68 \mid 64.5, 2.5),$$
$$P(X > 68 \mid Y = 1) = \mathrm{sf}(68 \mid 70, 3).$$

Compute values of survival functions using `scipy.stats`.

# Theorem

Suppose

$$X \sim N(\mu_0, \sigma_0^2),$$
$$Y \mid X = x \sim N(x, \sigma^2).$$

Then

$$X \mid Y = y \sim N(M_y, S^2),$$

where

$$M_y = \frac{\frac{1}{\sigma_0^2}\mu_0 + \frac{1}{\sigma^2}y}{\frac{1}{\sigma_0^2} + \frac{1}{\sigma^2}} \qquad \text{and} \qquad S^2 = \frac{1}{\frac{1}{\sigma_0^2} + \frac{1}{\sigma^2}}.$$

Suppose that intrinsic intelligence quotient (IQ) is normally distributed in the population with mean 100 and standard deviation 15.

Suppose, further, that IQ test scores for a person with an IQ of $x$ are normally distributed with mean $x$ and standard deviation 5.

What is the probability that a person who scores 120 on an IQ test has IQ at least 120?

Let $X$ denote intrinsic IQ and let $Y$ denote test score. Then

$$X \sim N(100, 15^2),$$
$$Y \mid X = 120 \sim N(120, 5^2).$$

By the Theorem,

$$X \mid Y = y \sim N(M_y, S^2),$$

where

$$M_y = 118 \qquad \text{and} \qquad S^2 = 22.5 = 4.74^2.$$

It follows that

$$P(X > 120 \mid Y = 120) = \mathrm{sf}(120 \mid 118, 4.74) = 0.34.$$